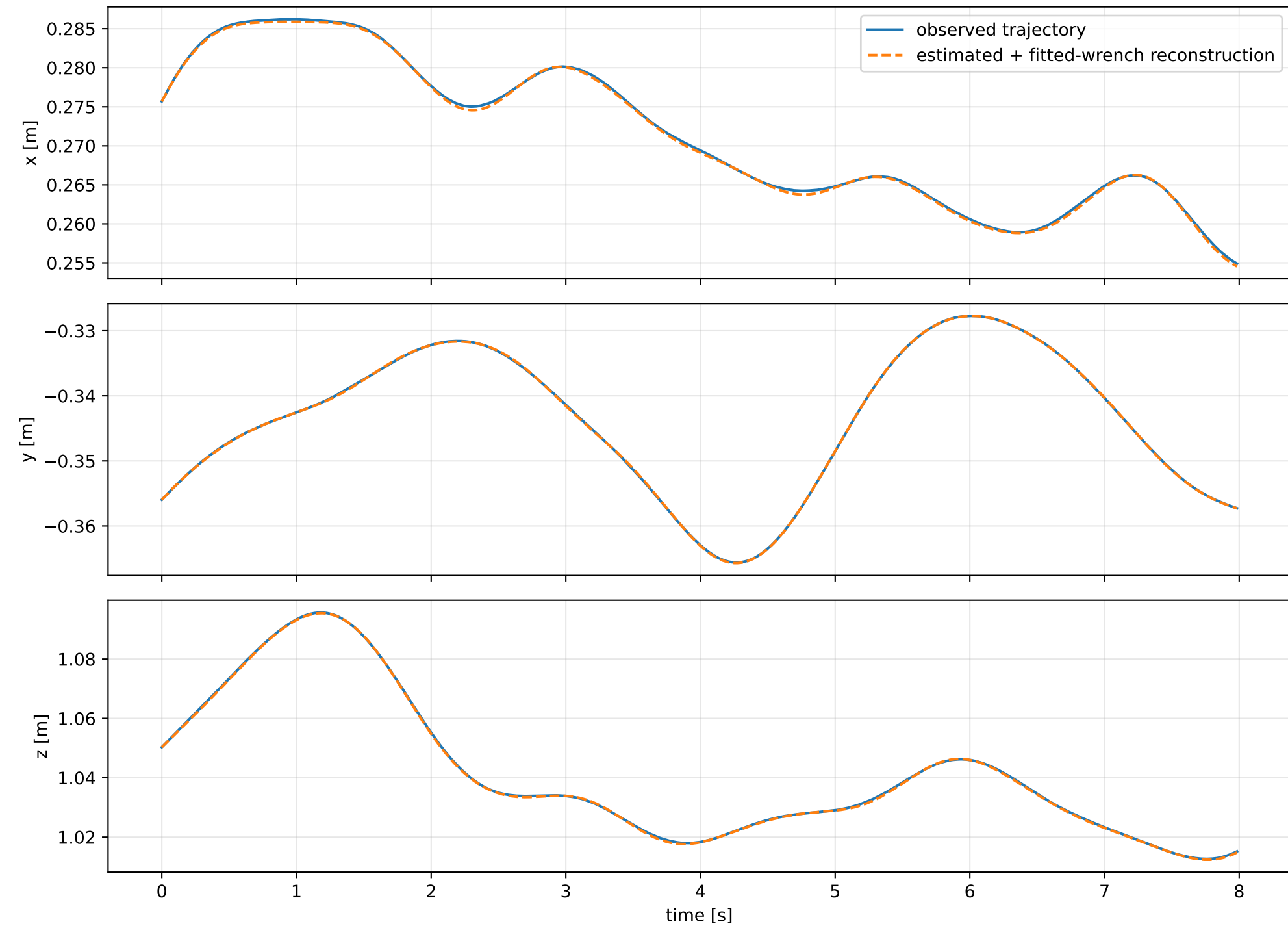
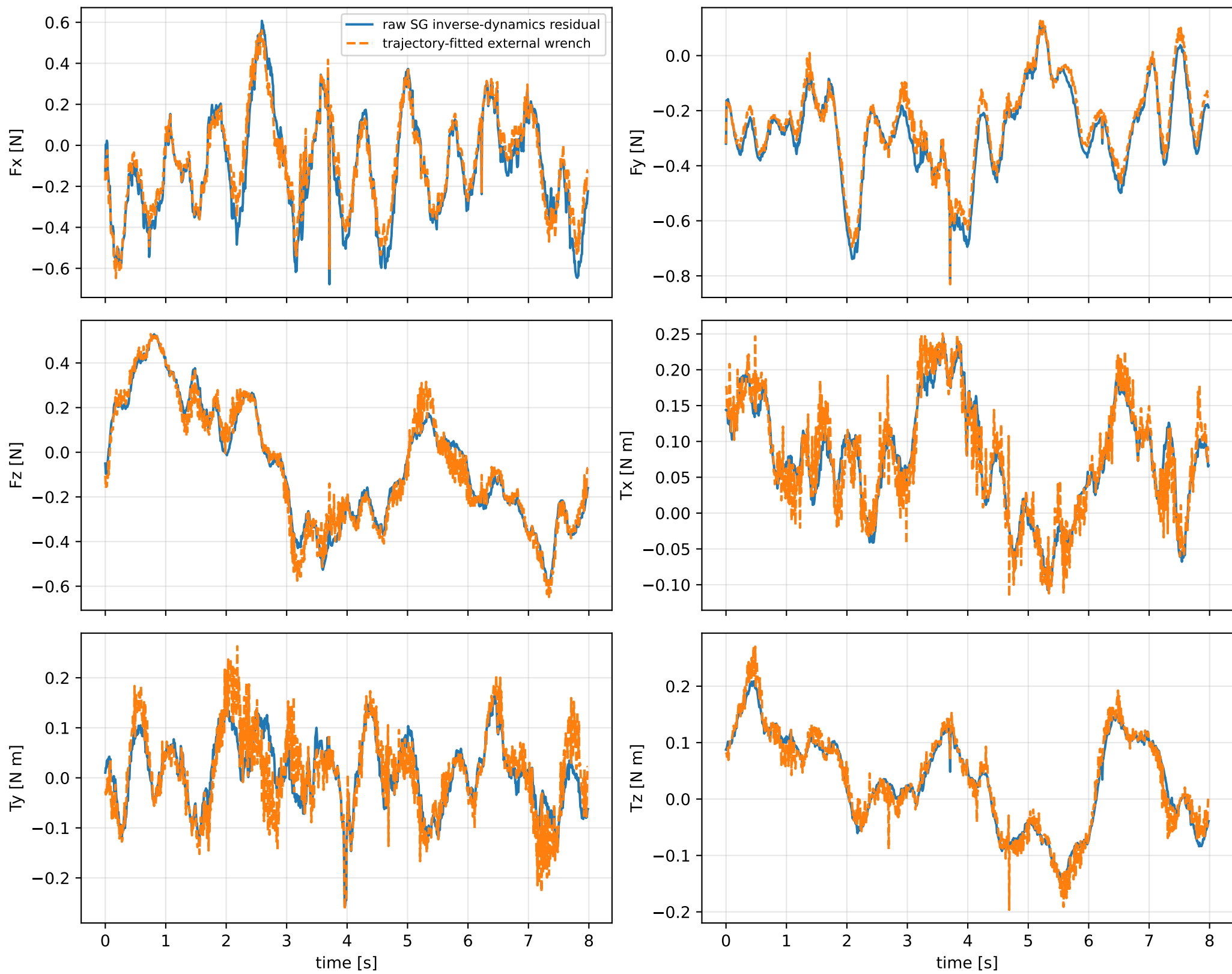


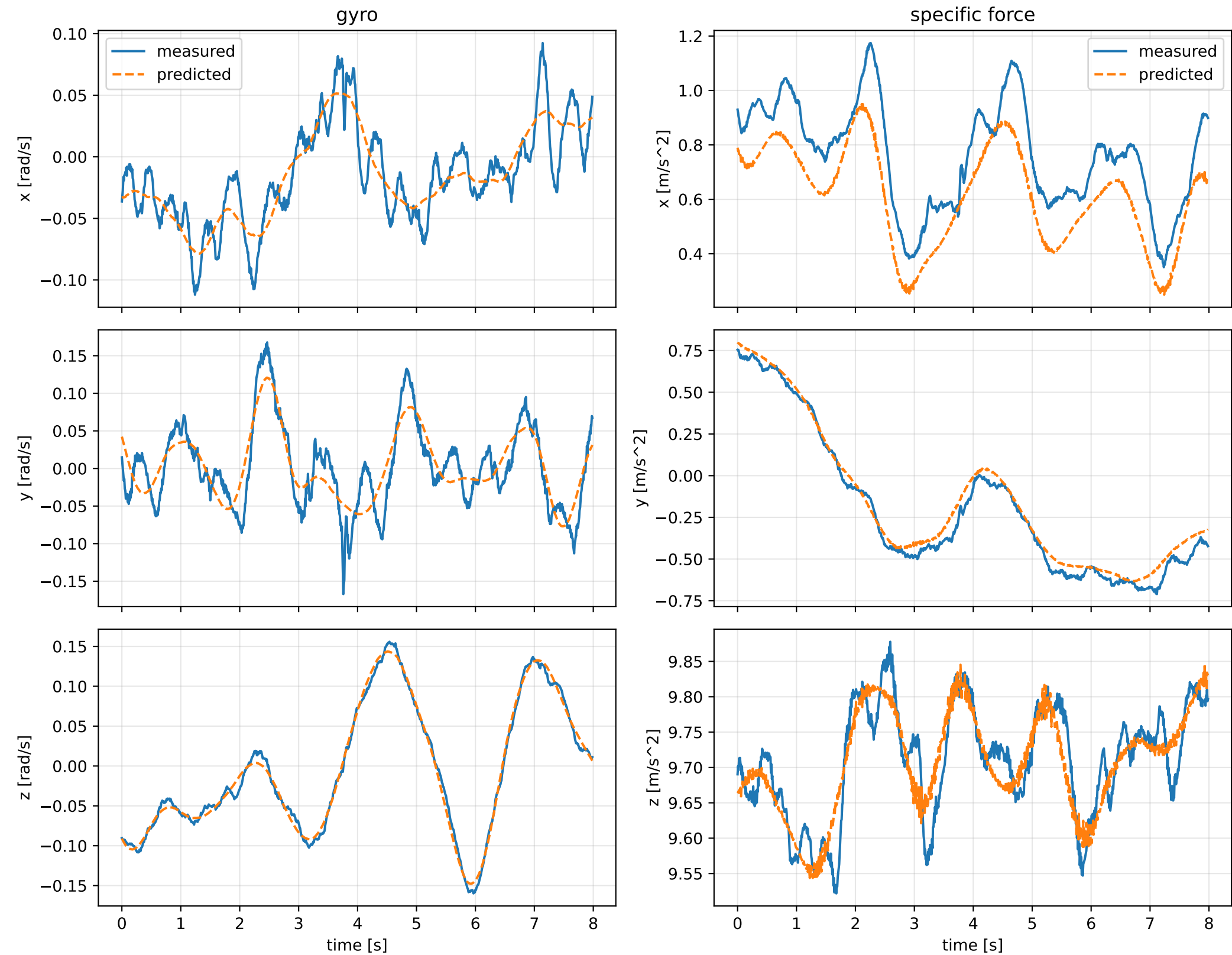
sweep_sg_window_covariance__004: trajectory



sweep_sg_window_covariance_004: residual wrench



sweep_sg_window_covariance_004: measured sensor comparison



sweep_sg_window_covariance__004: nominal -> estimated parameters

Case: sweep_sg_window_covariance__004
status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 4.18979451 delta 1.83819692

inertia [kg m^2] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [0.8697898 0.0864423 0.2524098] d=[0.8047898 0.0864431 0.2523908]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [0.0864423 0.6657278 0.0998116] d=[0.0864431 0.6007751 0.0998116]
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [0.2524098 0.0998116 0.5737338] d=[0.2523908 0.0998116 0.4447417]

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [-0.0349374 -0.0561614 0.0037604] d=[-0.0329127 -0.0561309 0.0037604]

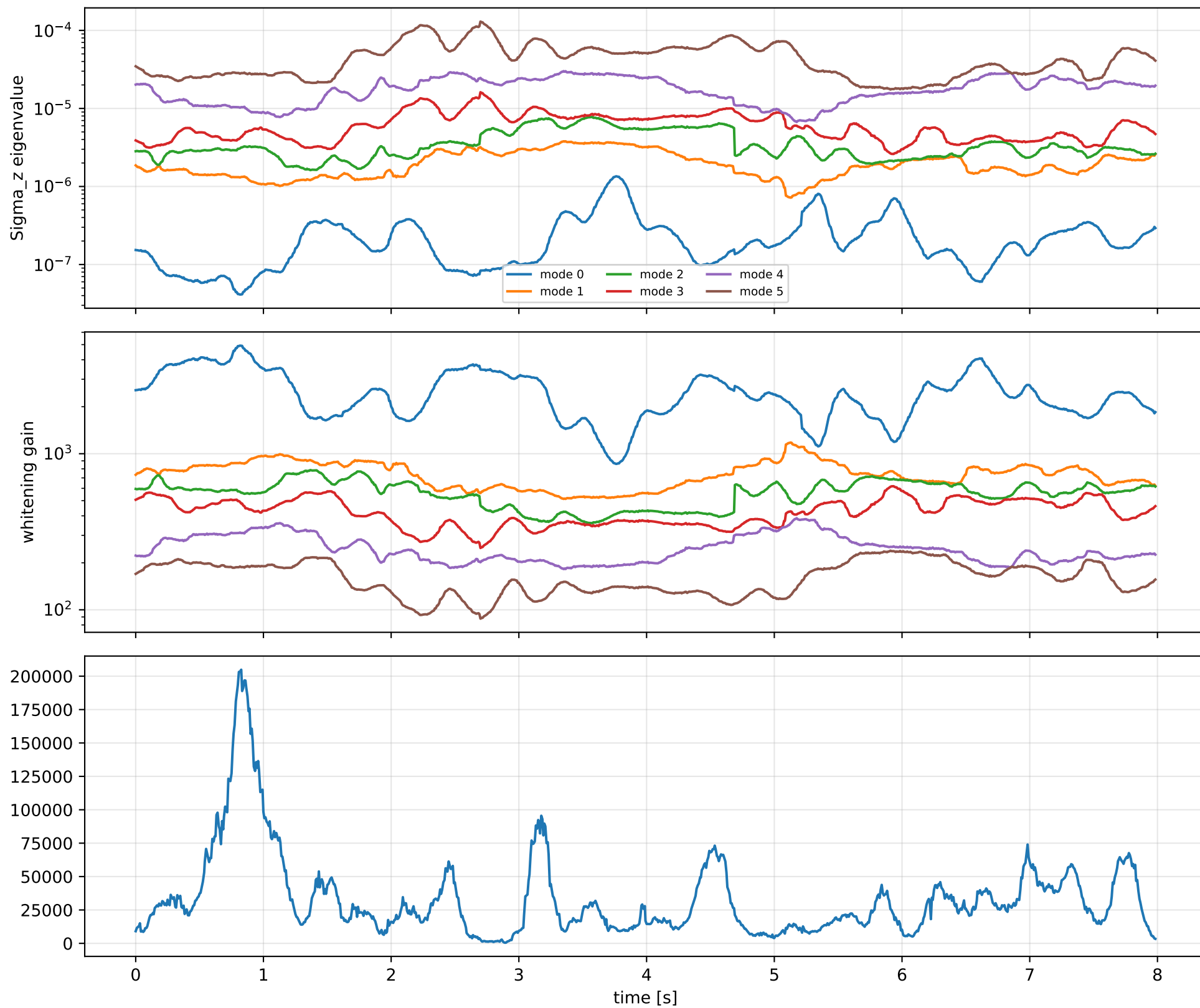
rotor force effectiveness

[1. 1. 1. 1.] -> [1.8567439 1.8767991 0.7058051 1.3676904] d=[0.8567439 0.8767991 -0.2941949 0.3676904]

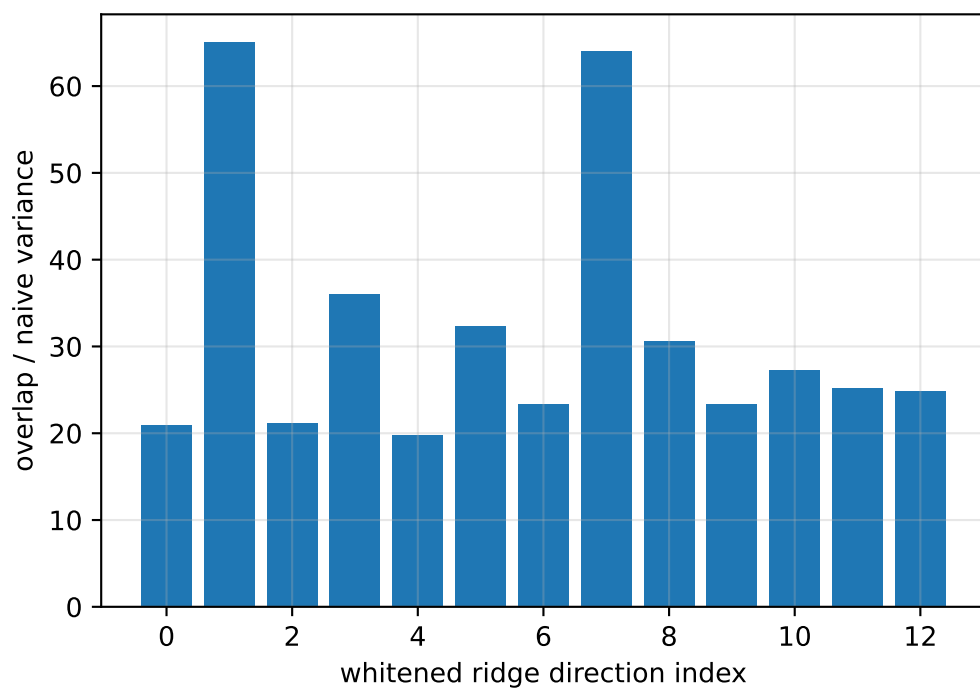
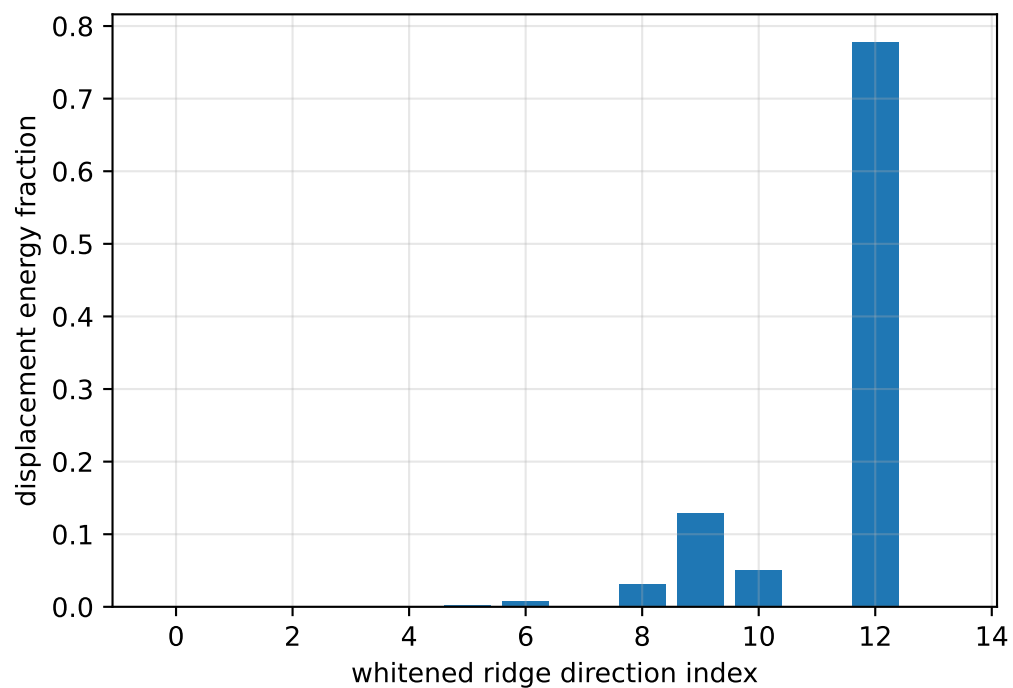
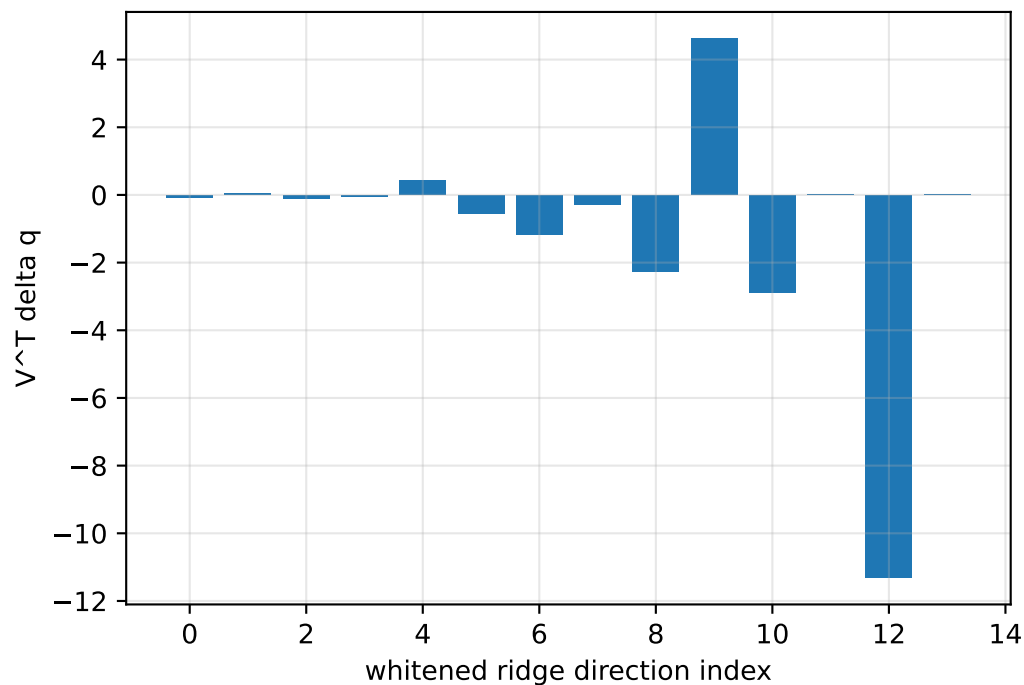
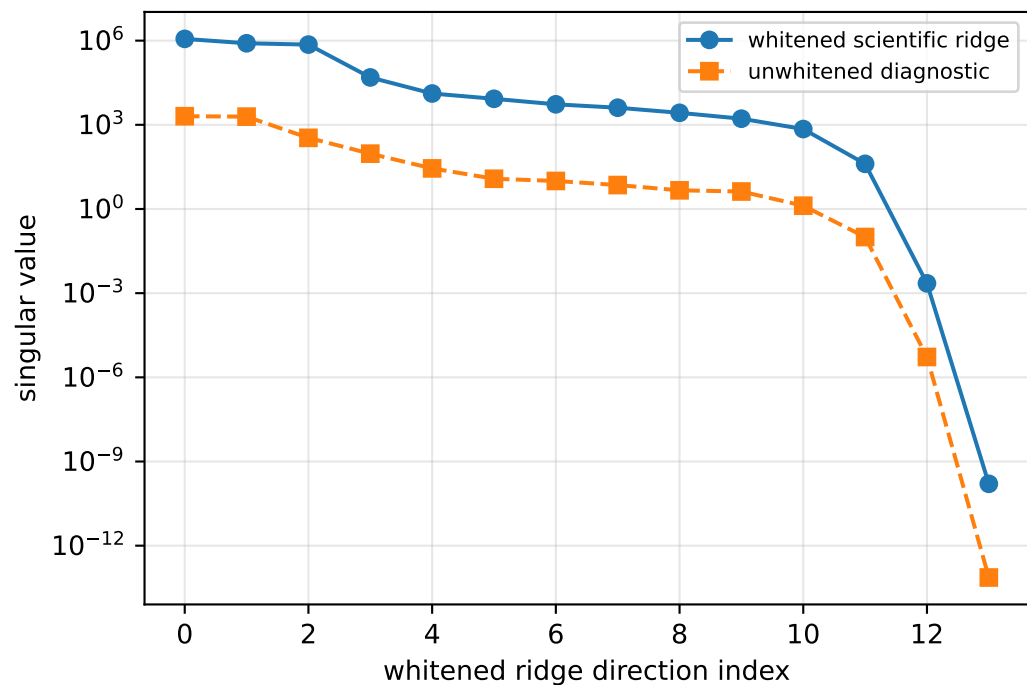
rotor lag [s] 0.2

gimbal lag [s] 2.19177642e-11

sweep_sg_window_covariance__004: covariance weighting

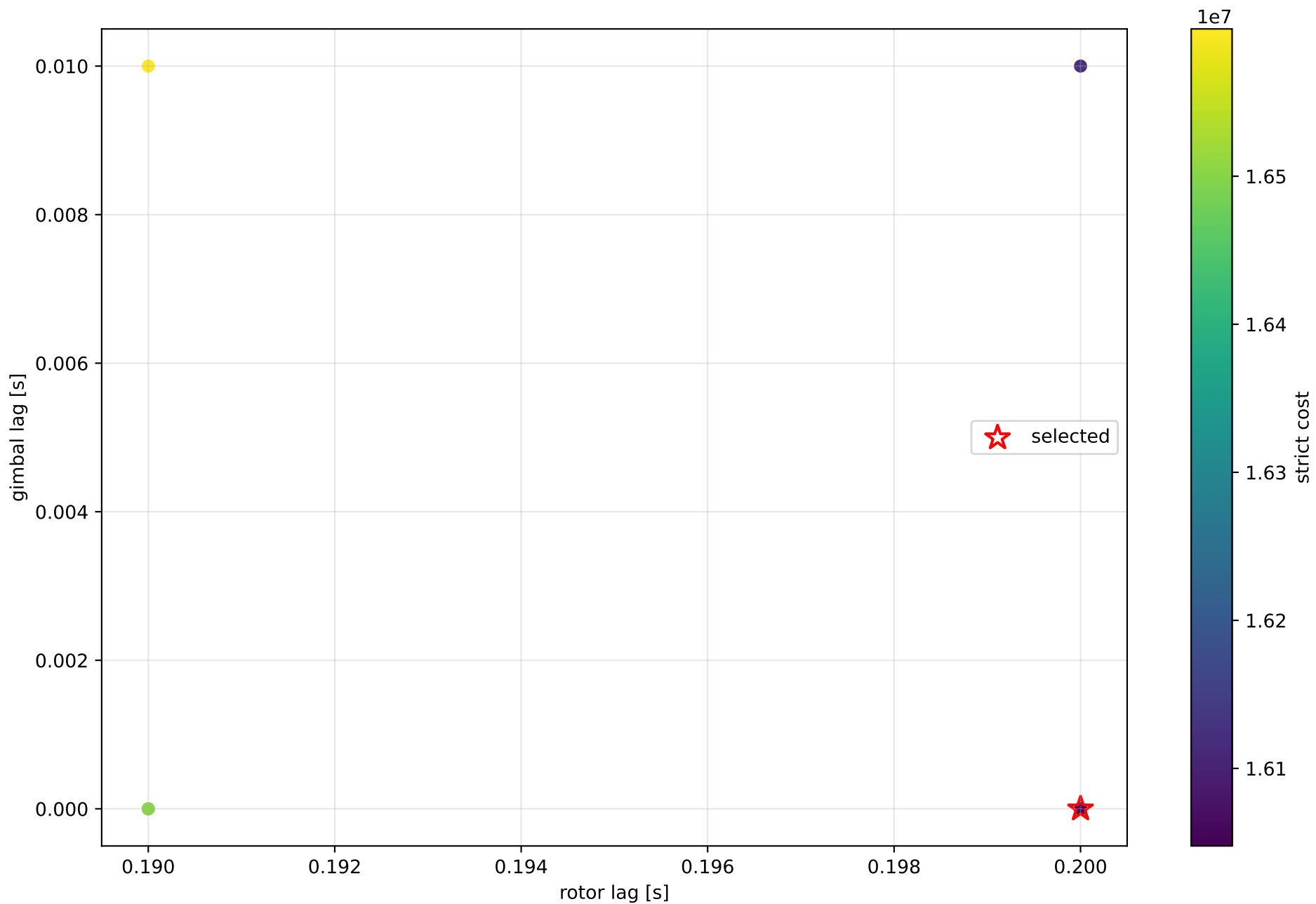


sweep_sg_window_covariance_004: parameter ridge



exact scale gauge: $\|v_{\text{scale}}\|=1.28036\text{e-}10$, rank=13, nullity=1

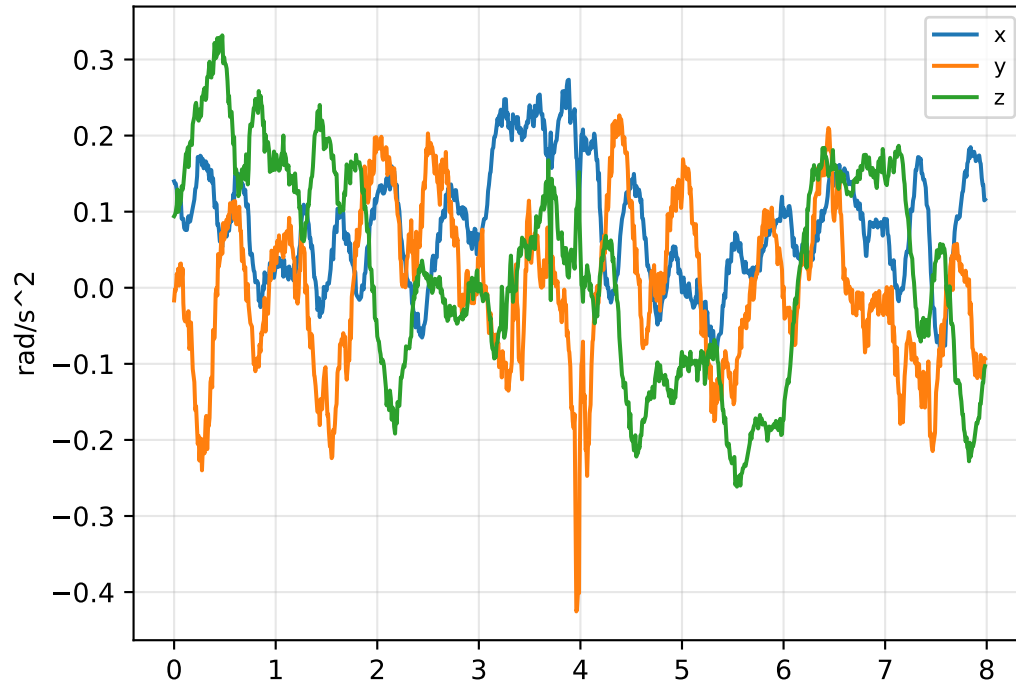
sweep_sg_window_covariance__004: strict lag candidates



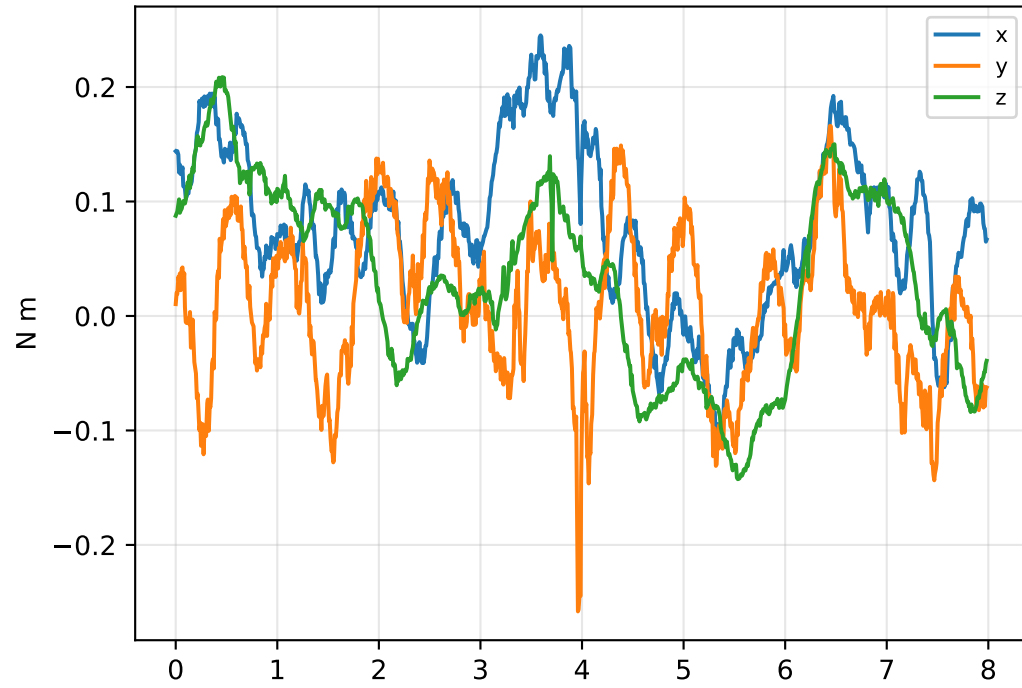
boundary flags: rotor lower=False upper=True; gimbal lower=False upper=False

sweep_sg_window_covariance_004: acceleration/wrench closure

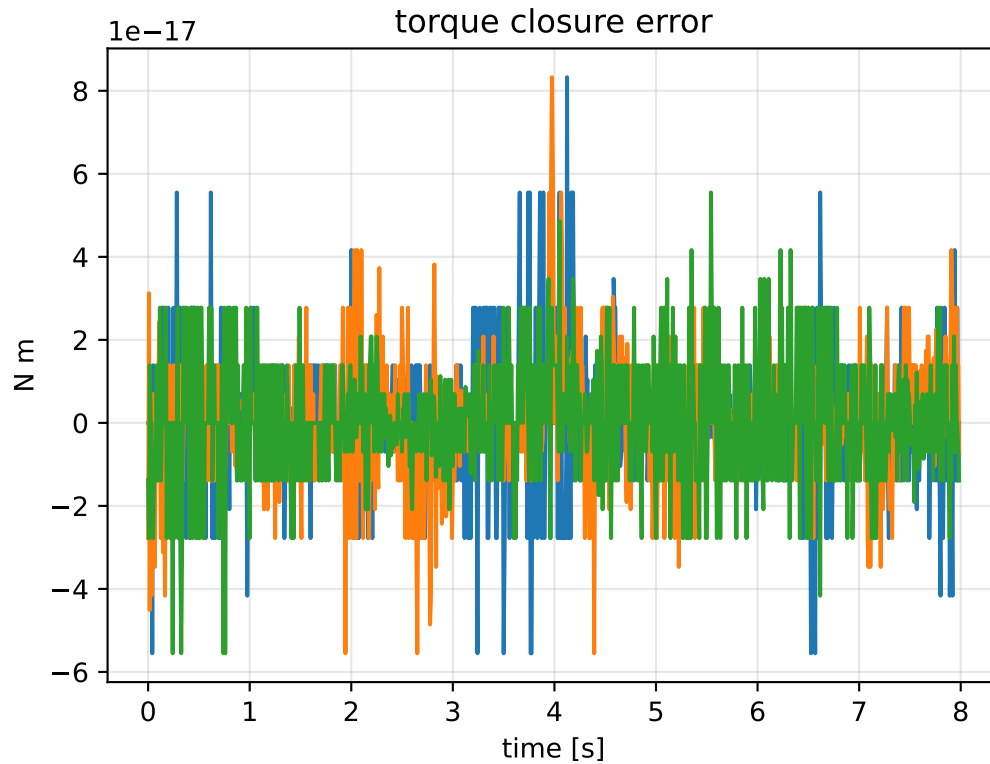
angular-acceleration residual



raw torque residual



torque closure error



principal moments $[\text{kg m}^2]$
 nominal: $[0.064953 \ 0.065 \ 0.128992]$
 estimated: $[0.420456 \ 0.634171 \ 1.054624]$
 estimated / nominal: $[6.473263 \ 9.756473 \ 8.175883]$

force closure max/rms: $1.25733e-14 / 2.45356e-15$
 torque closure max/rms: $8.32667e-17 / 1.4742e-17$